

Ideation Phase

Define the Problem Statement

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Project Name: Plugging into the Future: An Exploration of Electricity Consumption Patterns

Customer Problem Statement

Problem Statement (PS)	Description
I am (Customer)	An energy analyst, government official, researcher, utility manager, or decision-maker responsible for monitoring electricity consumption.
I'm trying to	Analyze electricity consumption patterns, compare electricity usage across states and regions, and support better planning.
But	The available data is scattered across spreadsheets and reports, making analysis difficult.
Because	Traditional reports lack interactive visualizations and require significant effort to interpret.
Which makes me feel	Frustrated, time-constrained, and uncertain while making decisions related to energy planning.

Final Problem Statement

Energy analysts, government agencies, researchers, and utility managers need an interactive platform to analyze electricity consumption because existing reports and raw datasets make it difficult to compare state-wise and region-wise electricity usage. This limits informed decision-making for energy planning. Therefore, this project proposes an interactive Tableau dashboard integrated with a Flask web application to provide meaningful visual insights and support effective energy management.

Proposed Solution

- Visualize electricity consumption across Indian states and regions.
- Compare electricity usage between 2019 and 2020.
- Display interactive maps, pie charts, and bar charts.
- Provide filters for easy exploration of data.
- Integrate the Tableau dashboard with a Flask web application.